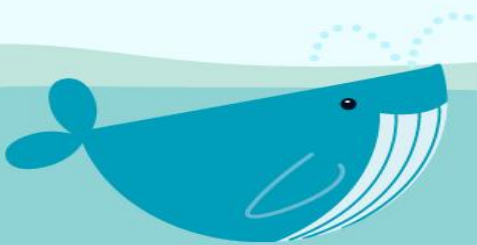




Humpback Whale Individual Identification

By Lisa Liu | Osvaldo Valdiviezo | Kenia Sanchez-Macario



Project Overview

Goal:

Automate identification of individual humpback whales using machine learning.

Motivation:

- Fluke patterns are unique like fingerprints
- Manual ID is slow and expert-dependent
- Automation supports large-scale marine research

Approach:

- Preprocess Kaggle dataset
- Train PCA+SVM, ANN, and CNN
- Compare accuracy, ROC curves, and AUC



Dataset Description

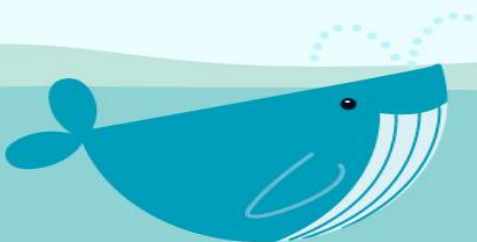
Source: Kaggle Humpback Whale Identification Competition

Filtered Dataset:

- Removed *new_whale* images
- Kept **top 20 whale IDs** with ≥ 20 images
- Converted all images to **96×96 grayscale**

Final Counts:

- ~600–700 images
- 20 classes
- 80% train / 20% validation



Data Challenges

Real-World Variability in Whale Photography

- Images captured by different researchers, boats, and cameras
- Wide variation in lighting (bright sun, overcast, glare on water)
- Motion blur from waves, splashes, and whale movement
- Flukes photographed at different angles and distances

Biological & Environmental Complexity

- Dorsal vs. ventral fluke views
- Natural markings vary in contrast and visibility
- Water spray and foam often obscure key features

Impact on Modeling

- Harder for models to learn consistent features
- Requires robust preprocessing and balanced training
- CNNs benefit most because they learn spatial patterns



Preprocessing Pipeline

Why Preprocessing Matters

Humpback fluke images vary dramatically in size, color, and orientation. Standardizing inputs ensures all models learn from consistent data.

Flat-Input Models (SVM & ANN)

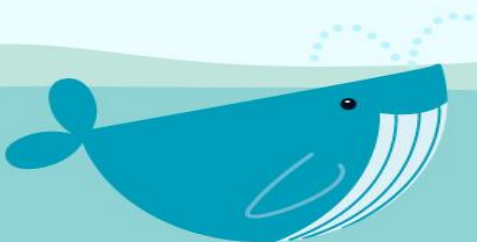
- Resize to **96×96** using LANCZOS (high-quality resampling)
- Convert to **grayscale** to reduce noise and dimensionality
- Flatten to **9,216-dim vectors**
- Normalize pixel values to **[0,1]**
- Apply **StandardScaler** (fit on train only)
- PCA used later to reduce dimensionality for SVM

CNN Pipeline (Spatial Input)

- Resize to **96×96**
- Convert to grayscale → shape **(96, 96, 1)**
- Normalize to **[0,1]**
- Preserve spatial structure for convolutional filters

Why Two Pipelines?

- SVM/ANN require flat vectors
- CNN requires spatial layout to detect edges, curves, and textures



Train/Validation Strategy

Stratified 80/20 Split

- Ensures each whale ID appears proportionally in both sets
- Prevents rare classes from being underrepresented in validation

Balanced Class Weights

- Some whales have 3× more images than others
- Class weights prevent the model from favoring majority classes
- Especially important for SVM and ANN

Why This Matters

- Small dataset → high risk of overfitting
- Balanced and stratified splits improve fairness and stability

Random State = 5

- Ensures reproducibility
- Allows fair comparison across all three models



Model 1: PCA + SVM

Architecture / Pipeline

- Flattened 96×96 images → 9,216 features
- PCA reduces to **300 components** (90% variance retained)
- RBF-kernel SVM classifier

Strengths

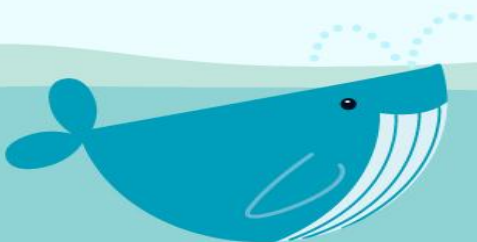
- Strong baseline for high-dimensional data
- PCA reduces noise and redundancy
- Performs well even with small datasets

Training Setup

- GridSearchCV over:
 - **C**: {1, 10, 100, 1000}
 - **gamma**: {scale, 0.01, 0.001, 0.0001}
- Balanced class weights
- Fastest training time among all models

Limitations

- Flattening destroys spatial structure
- Cannot learn texture or shape patterns
- Accuracy capped by PCA representation



Model 2: Keras ANN

Architecture

- Input layer: **9,216 units**
- Dense 512 (ReLU)
- Dense 128 (ReLU)
- Output: Softmax(20 classes)

Training Setup

- Adam optimizer
- Sparse categorical cross-entropy
- GridSearchCV over:
 - **batch_size**: {16, 32}
 - **epochs**: {30, 50}
- Medium training time

Strengths

- Learns nonlinear decision boundaries
- Simple architecture, easy to tune
- Performs reasonably well on structured data

Limitations

- Flat input discards spatial relationships
- Underperforms SVM and CNN
- Requires more data to generalize well



Model 3: Custom CNN

Architecture

- 4 convolutional blocks:
 - Filters: **32** → **64** → **128** → **256**
 - Each block: Conv → BatchNorm → ReLU → MaxPool → Dropout
- Dense 512 + Dropout
- Output: Softmax(20 classes)

Training Setup

- Adam (lr = 1e-3)
- EarlyStopping (patience = 12)
- ReduceLROnPlateau (factor = 0.3, patience = 4)
- Up to 80 epochs, batch size 16
- Slowest training time (GPU recommended)

Strengths

- Learns hierarchical spatial features
- Best accuracy and AUC
- Dropout + BatchNorm improve generalization

Limitations

- Computationally expensive
- Requires more data for peak performance
- Sensitive to hyperparameters



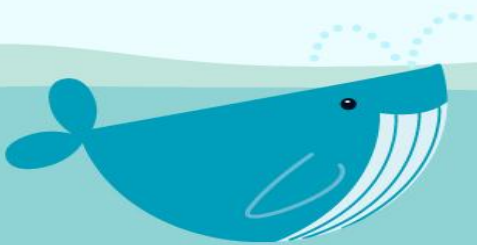
Model Performance Comparison

Validation Results:

- **PCA + SVM:** 55–70% accuracy, AUC 0.87–0.93
- **ANN:** 45–65% accuracy, AUC 0.82–0.90
- **CNN:** 65–80% accuracy, **AUC 0.90–0.96**

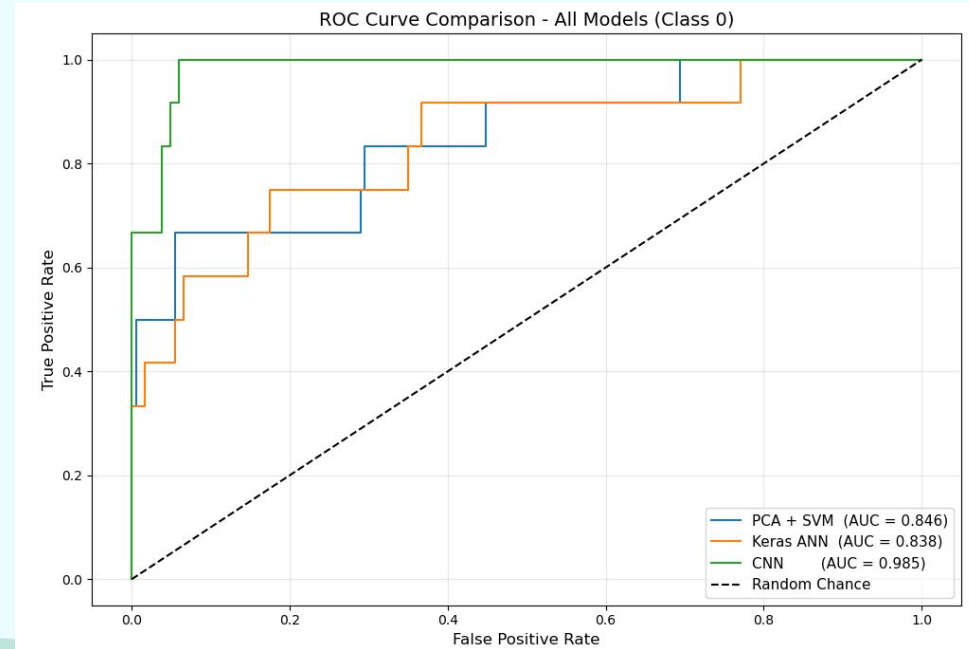
Conclusion:

CNN performs best due to spatial feature learning. The results show a clear trend: **models that preserve spatial structure consistently outperform flat-input approaches**



ROC Curve Comparison

- One-vs-Rest ROC curves generated for all 20 classes
- CNN shows highest TPR at low FPR
- SVM competitive but limited by PCA
- ANN weakest separation



Key Findings

1. CNNs Capture Critical Spatial Features

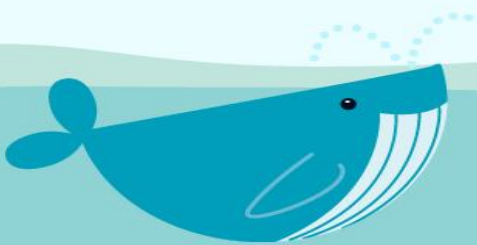
The CNN achieved the highest accuracy and AUC because it learns edges, textures, and fluke patterns that flat models cannot represent.

2. PCA + SVM Is a Strong Baseline

Despite losing spatial structure, PCA-reduced SVM performed competitively and provided a reliable, efficient benchmark.

3. Dataset Size Limits All Models

With ~600–700 images and class imbalance, all models face generalization challenges, highlighting the need for augmentation and transfer learning.



Limitations

1. Limited Dataset Size

Only ~600–700 images across 20 classes restrict model generalization and increase overfitting risk.

2. Class Imbalance

Some whale IDs have 3× more images than others, making it harder for models to learn minority classes.

3. No Data Augmentation

Without flips, rotations, or brightness adjustments, the effective dataset size remains small.

4. No Transfer Learning

Using only custom models limits performance compared to pre-trained CNNs like ResNet or EfficientNet.



Future Work

1. Add Data Augmentation

Increase effective dataset size with flips, rotations, cropping, and brightness jitter.

2. Use Pre-trained CNNs

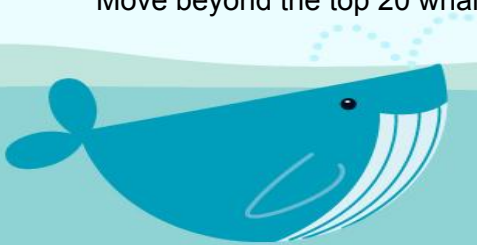
Leverage models like ResNet50 or EfficientNet to improve accuracy with limited data.

3. Explore Metric Learning

Siamese networks or triplet loss can better capture fine-grained differences between whale flukes.

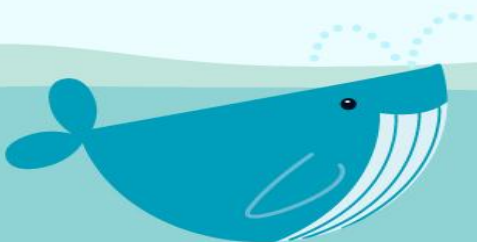
4. Scale to Full Dataset

Move beyond the top 20 whale IDs toward the full 5,000+ ID challenge.



Team Responsibilities

Lisa Liu	Osvaldo Valdiviezo	Kenia Sanchez-Macario
Model Training	Data Analysis	Data Analysis
Documentation	CNN Training	Documentation
Peer Review	Documentation	Presentation



Thank You

Questions?

